41. DEVELOPMENT OF THE RIBS

DURATION : 07:22 STUDY SHEET

COURSE SUMMARY

This video delves into the embryological development of the ribs, linking elements such as the ectoderm and mesoderm to the formation of thoracic structures. It expounds on the concepts of complexity and emergence, and demonstrates how these processes can influence clinical conditions. In parallel, the importance of rib motility is highlighted, offering both direct and indirect techniques for working on each rib. The emotional aspect, particularly in relation to the fourth rib on the left, is explored, underscoring the correlation between respiratory movements and emotional balance.

RIB DEVELOPMENT

The **ribs** are linked to a complex developmental process. This development involves the **ectoderm**, **mesoderm**, **vascular anatomy**, **cerebralisation**, **segmentation**, as well as the development of the **gut**, **cavities**, **heart**, and **liver**. All these elements contribute to the overall emergence of the costal system.

This phenomenon is often described by **emergence theories**, which suggest that a simple process can give rise to more complex structures. At a certain stage, complexity increases, but there comes a time when the system can no longer absorb this complexity. It can then return to a state of **chaos** to rebuild itself. This process of deconstruction and reconstruction can be difficult to experience, just as **depression** can occur after a period of high pressure.

The ribs form as **densifications** between the segmented thoracic vessels, beginning around day 35 of development. The vertebral body and the growing rib arch lengthen, following the developmental movement of the embryo in its **amniotic cavity**. This process of **cephalisation**, **cardialisation**, **hepatisation**, and **delimitation** is essential for rib development.

From day 45, a **fracture** occurs, giving rise to a **chondrocostal** joint that progressively centres towards the midline, at the level of the **clavicle**. The sternal bars, which are mesenchymal condensations, also join at the angle of the ear, corresponding to the **mediastinum**.

After birth, the **caecum** completes its descent, and the embryo continues to develop in a similar dynamic. Structures around the vertebra, such as the **myotomes**, **dermatomes**, and **sclerotome**, gradually form, at the same time as the process of embryonic delimitation.

The ribs are also linked to clinical phenomena. For example, in boxers, a blow can lead to **intercostal evisceration**. If the rib is not broken, this may indicate an underlying problem. In parallel, rib development is accompanied by the **parietal pleura** and the internal fold of the **visceral pleura**, forming the **endothoracic fascia**.

Working on rib motility is essential. By deepening this approach, a connection can be established between the parietal pleura and the visceral pleura, particularly at the level of the **pulmonary hilum**. Freeing the ribs allows the lung to be freed, thus promoting essential exchanges.

It is possible to work on each rib individually or to adopt a more global approach using direct or indirect techniques. A free costal cage is crucial to allow a fluid circulation of exchanges.

An important point to note is the emotional impact of the **fourth rib on the left**, which is often associated with phenomena of anxiety and **precordialgia**. The ribs work in synchronisation with **inhalation** and **exhalation**. If a rib does not descend correctly, this can lead to emotional blockages.

Working on this area involves bringing the rib back into its inhalation and exhalation movement, correcting its position and balancing the pulmonary hilum. This process is essential to restore the balance and functionality of the costal system.

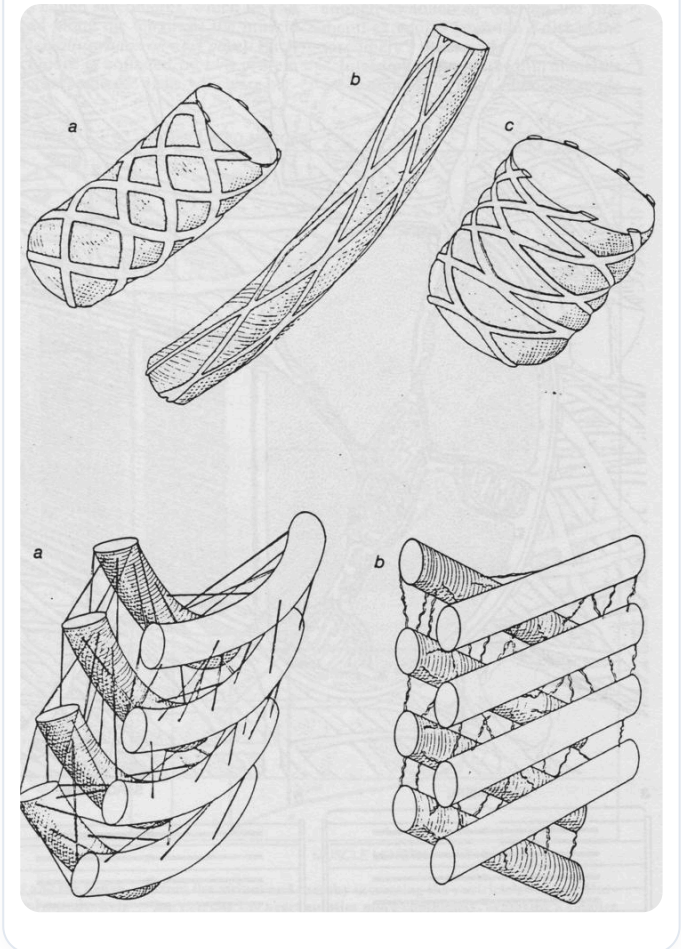


FIG. — Tensegrity

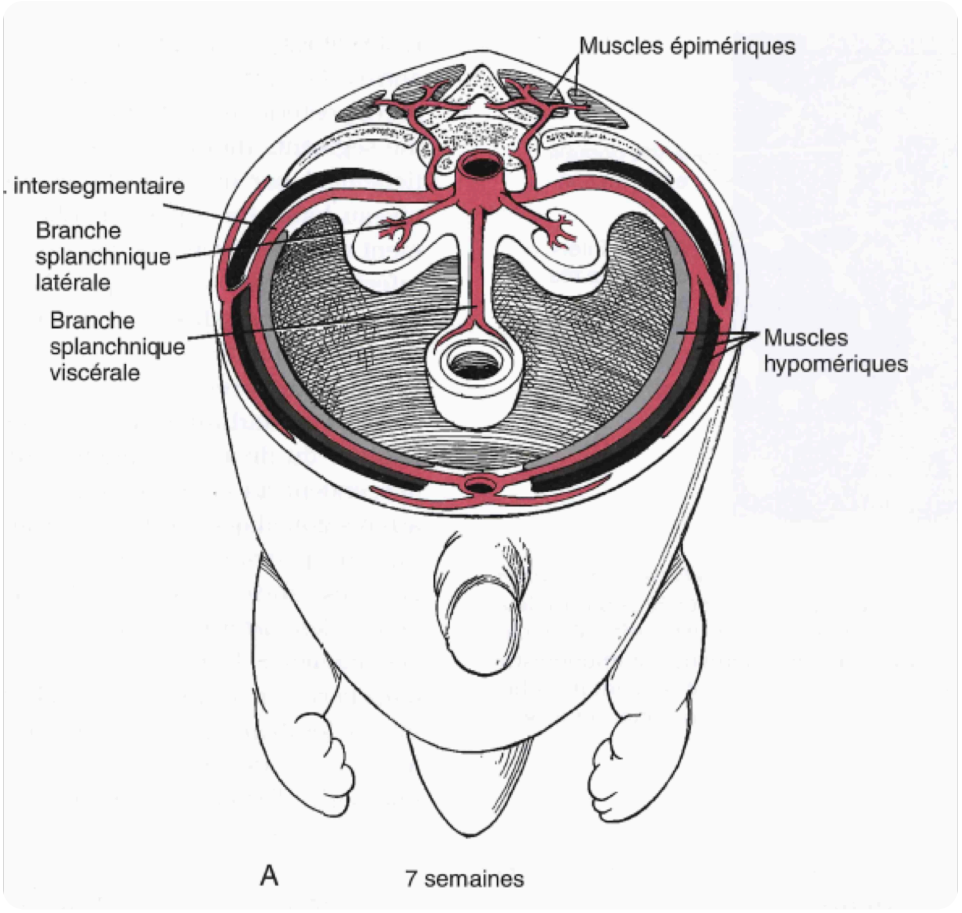


FIG. — Embryo 7 weeks, vessels

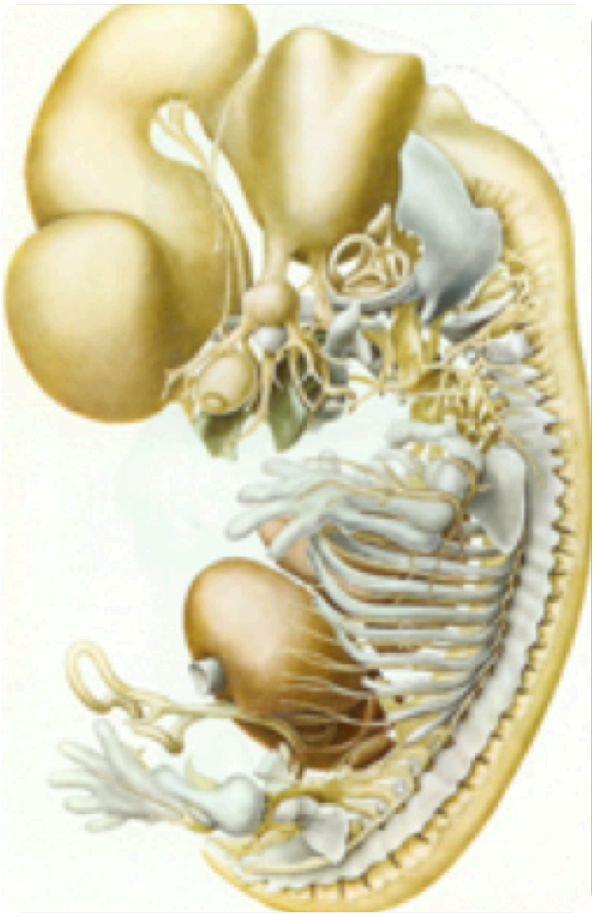


FIG. — Embryo nervous system

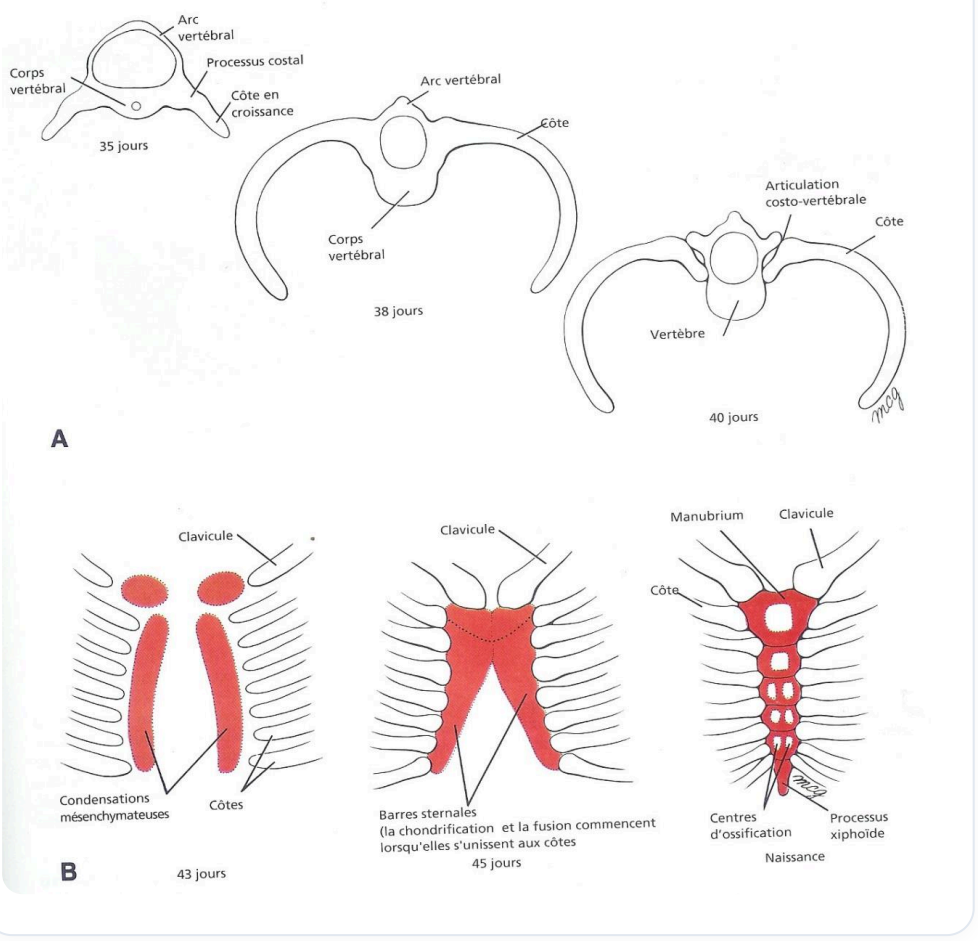


FIG. — Development of ribs and sternum